

Parametric cerebral blood flow and arterial transit time mapping using a 3D convolutional neural network

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Purpose: To reduce the total scan time of multiple postlabeling delay (multi-PLD) pseudo-continuous arterial spin labeling (pCASL) by developing a hierarchically structured 3D convolutional neural network (H-CNN) that estimates the arterial transit time (ATT) and cerebral blood flow (CBF) maps from the reduced number of PLDs as well as averages.

Methods: A total of 48 subjects (38 females and 10 males), aged 56–80 years, comprising a training group (n = 45) and a validation group (n = 3) underwent MRI including multi-PLD pCASL. We proposed an H-CNN to estimate the ATT and CBF maps using a reduced number of PLDs and a separately reduced number of averages. The proposed method was compared with a conventional nonlinear model fitting method using the mean absolute error (MAE).

Results: The H-CNN provided the MAEs of 32.69 ms for ATT and 3.32 mL/100 g/min for CBF estimations using a full data set that contains six PLDs and six averages in the 3 test subjects. The H-CNN also showed that the smaller number of PLDs can be used to estimate both ATT and CBF without significant discrepancy from the reference (MAEs of 231.45 ms for ATT and 9.80 mL/100 g/min for CBF using three of six PLDs).

Conclusion: The proposed machine learning-based ATT and CBF mapping offers substantially reduced scan time of multi-PLD pCASL.

KEYWORDS

arterial spin labeling, arterial transit time, cerebral blood flow, convolutional neural network, deep learning

1 | INTRODUCTION

Arterial spin labeling (ASL) is a noninvasive MRI technique to measure cerebral blood flow (CBF). ASL uses magnetically labeled water protons in arterial blood as

a perfusion tracer.^{1–3} After a certain amount of time to allow the labeled arterial blood to be delivered to its final exchanging location in the brain tissue, the local CBF can be visualized. To quantify brain perfusion in ASL, labeled and nonlabeled images are acquired. Subtraction

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of acquired labeled image and nonlabeled control image cancels out static brain tissue signals and visualizes brain perfusion. The time required for the labeled blood to travel from the labeling location to its final location is defined as arterial transit time (ATT). The transit time varies by the location of the brain. Therefore, a single postlabeling delay (PLD) is not ideal in estimating the CBF of each voxel of the brain. Moreover, the selection of a single PLD is quite challenging, as ATT can vary among individuals.⁴⁻⁶ In this respect, multi-PLD ASL methods have been used widely for ASL studies. Additionally, an existing multi-PLD integrated pseudo-continuous ASL (pCASL) study has shown that the use of a variable bolus width and variable TR resulted in improved estimation accuracy of ATT and CBF with time efficiency.⁴ The proposed method outperforms any other conventional multi-PLD pCASL methods regarding estimation accuracy. However, the acquisition time is still relatively long due to the implementation of multiple PLDs. The relatively long scan time of multi-PLD pCASL limits its clinical accessibility. Thus, if the total scan time of multi-PLD pCASL can be reduced, the clinical accessibility of multi-PLD ASL will be increased.

Recent medical imaging studies have shown that machine learning algorithms can be used to improve the viability or accessibility of various medical imaging modalities. Using machine learning-based approaches enabled the classification or prediction of the diagnosis of specific diseases.⁷⁻⁹ The undersampled data set could be recovered from only a few features given by information redundancy using deep learning.^{10,11} Deep learning-based approaches have also offered advantages in ASL studies. Specifically, deep learning-based methods have been used to denoise ASL and provide a higher SNR.¹²⁻¹⁴ Another study used a convolutional neural network (CNN) model to improve the perfusion-weighted image (PWI) by estimating PWI from a smaller number of averages.¹⁵ In this respect, the implementation of a CNN can potentially recover the undersampled multi-PLD pCASL perfusion map. In a more recent study, researchers showed the feasibility of estimating ATT and CBF maps from multi-PLD pCASL using U-Net and CNN.¹⁶

In this study, we developed a 3D hierarchical CNN (H-CNN) model to estimate ATT and CBF maps at different depths. The proposed technique will allow a total scan time reduction of multi-PLD pCASL acquisition by recovering the reduced PLD information or the reduced averages. In addition, this study may provide ideal selections of PLDs by comparing every single combination of PLDs using both nonlinear model fitting and deep learning-based methods. Thus, our H-CNN model will eventually increase the accessibility of the multi-PLD pCASL perfusion and ATT maps in a clinical setting.

2 | METHODS

2.1 | Ethics

This work involved human subjects. Approval of all ethical and experimental protocols was granted by the Wake Forest University Health Sciences Institutional Review Board Office. The participants provided their written informed consents to participate in this study.

2.2 | Multi-PLD pCASL

A single PLD ASL is used widely in conventional ASL settings to create PWIs. However, due to its limitations related to the single delay, multi-PLD ASL with varying TI and fixed long TR is also widely used. In pCASL, TI is the summation of labeling duration and PLD. Recently, a multi-PLD integrated pCASL scheme was proposed, which acquires images with varying TI and TR.⁴ Because of the varying TR, the proposed acquisition scheme allowed total scan-time reduction compared with using a long, fixed TR. In this study, we implemented the multi-PLD integrated pCASL scheme with varying TI and TR, excluding the very short PLDs used in the original work.⁴ To avoid a fixed TR dependence, the implementation of presaturations was required. The presaturation pulse allowed the tissue signal to recover from zero by saturating the imaging slice immediately before the labeling pulse.⁴ The tissue signal recovery from zero at the beginning of each TR removed the dependence of the signal on TR, which allowed TR to be varied according to each TI of the multi-PLD pCASL scheme.⁴ Furthermore, the presaturation pulse enabled us to use interleaved longitudinal magnetization, which is less motion sensitive. Figure 1A shows the interleaved longitudinal magnetization scheme.

2.3 | MRI acquisition

Forty-eight subjects were enrolled in the Wake Forest Alzheimer's Disease Research Center clinical core cohort and underwent MRI, including the multi-PLD pCASL scheme, on a 3T Siemens Skyra MRI with a 32-channel head coil (Siemens Healthineers, Erlangen, Germany). The subjects included 10 males and 38 females. The subjects had no severe neurological diseases such as stroke or dementia. The mean and SD of the age of the subjects were 67.31 and 6.8 years, respectively. T₁-weighted structural images with full brain coverage were obtained using a 3D MPRAGE sequence with an isotropic spatial resolution of 1 × 1 × 1 mm (TR = 2300 ms, TE = 2.98 ms, TI = 900 ms, flip angle = 9°, FOV = 240 × 256). For the

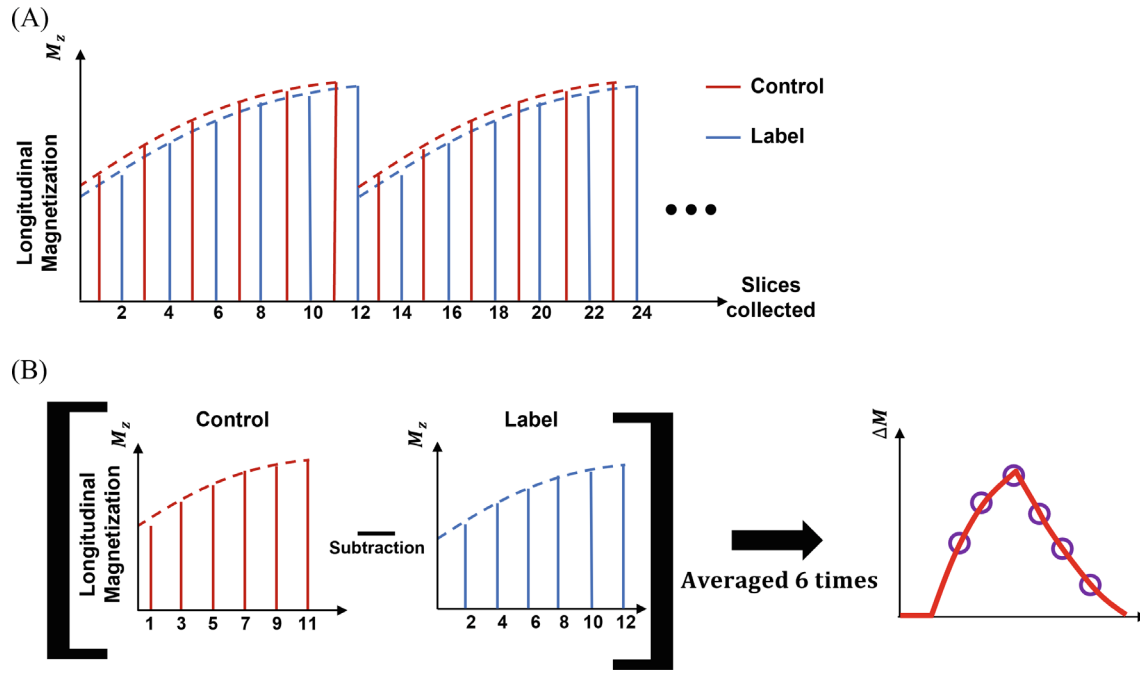


FIGURE 1 (A) Interleaved multi-postlabeling delay (PLD) pseudo-continuous arterial spin labeling (pCASL) longitudinal acquisition scheme. One label and one control image were acquired for each PLD. A total of six paired acquisitions were acquired. The presaturation pulse was applied immediately before each train of labeling pulse. (B) Theoretical description of subtraction of label and control images. Pairwise subtraction between the label and control slices results in an ASL standard curve.

multi-PLD pCASL scheme, a total of six PLDs from 0 to 3000 ms with increments of 600 ms and six averages per PLD were collected. The duration of pCASL labeling was 1800 ms. To use 0 labeling delay, the labeling duration was set to 1700 ms for the shortest TI only to allow time for the fat saturation pulse and a half of the excitation RF pulses. This is because it is not possible to acquire the image immediately following the 1800-ms label. Each TI had minimum TR (2900, 3500, 4100, 4700, 5300, and 5900 ms for the six TIs, respectively). The scan time for all six PLDs was 5 min 21 s. A 2D single-shot EPI acquisition was used to cover the whole brain (matrix size = $56 \times 70 \times 36$, resolution = $3 \times 3 \times 4$ mm), and slices were acquired in ascending order. The delay between slices was 27.5 ms. An M_0 map was acquired from a separate volume with EPI acquisition using identical resolution and parameters with 18-s TR. Flow crusher gradients on all three orthogonal axes were played immediately after an excitation RF pulse to remove flowing signals faster than 10 cm/s.

2.4 | Data preprocessing

Before ATT and CBF estimation, raw pCASL data were first realigned to the M_0 volume, which presents equilibrium magnetization using *SPM12*.¹⁷ The raw images were then smoothed with a 3D 4.5-mm Gaussian kernel.

Pairwise subtraction between the label and control volumes and averaging six repetitions resulted in a 4D PWI for each subject. The binary brain mask was created using the structural T_1 -weighted image with *SPM12*.¹⁷

2.5 | Ground-truth references: Nonlinear model fitting

A voxel-wise nonlinear fitting method was used to create ATT and CBF maps.⁴ A multivariable optimization method (“fminsearch” in *MATLAB*) was used on the acquired multi-PLD data using an ASL kinetic model.^{4,18} Note that ATT and CBF maps were estimated simultaneously in the nonlinear model fitting. However, ATT and CBF were fitted sequentially at the algorithm level. The pair-wise subtraction between control and label slices provided the ASL standard kinetic model (Figure 1B), which complies with the following equation¹⁸:

$$\text{CBF} = \Delta M / [2M_{0\text{blood}} T_{1\text{tiss}} \alpha \exp(-\text{ATT}/T_{1\text{blood}}) \cdot (1 - \exp(-(\text{TI} - \text{ATT})/T_{1\text{tiss}}))] \quad \text{for TI} < \text{ATT}$$

$$\text{CBF} = \Delta M / [2M_{0\text{blood}} T_{1\text{tiss}} \alpha \exp(-\text{ATT}/T_{1\text{blood}}) \cdot (\exp(-(\text{TI} - \text{ATT} - \tau)/T_{1\text{tiss}}))(1 - \exp(-\tau/T_{1\text{tiss}}))] \quad \text{for TI} > \text{ATT} \quad (1)$$

where ΔM denotes the magnetization difference between control and label voxels; $M_{0\text{blood}}$ is the equilibrium magnetization of the blood; α is the labeling efficiency; τ is the labeling duration; and $T_{1\text{blood}}$ and $T_{1\text{tiss}}$ are T_1 of blood and tissue, respectively. The $M_{0\text{blood}}$ was estimated from the M_0 of the CSF region ($M_{0\text{csf}}$), which was taken from the separately acquired M_0 map. The measured $M_{0\text{csf}}$ value was converted to $M_{0\text{blood}}$ using the measured CSF signal and assumed T_2^* relaxation times of blood and CSF.¹⁹ A fixed $M_{0\text{blood}}$ value was assumed because all data were collected with a single scanner in this study, but the estimated CBF images need to be accounted with the scanner-specific $M_{0\text{blood}}$ value if multiple scanners are used in a study. The labeling efficiency, α , was 0.95. $T_{1\text{blood}}$ (1660 ms) from literature²⁰ and was used for the experiment; $T_{1\text{tiss}}$ was assumed the same as the $T_{1\text{blood}}$ in this study. In the nonlinear fitting model, the PWIs were used as ΔM , and the maximum value in the perfusion time series was selected as an initial estimate. The calculated ATT and CBF maps from the nonlinear fitting method with a full data set, which contains six PLDs and six averages, were used for the ground-truth reference images for each subject.

2.6 | CNN

In brain imaging studies, the intrinsic spatial correlation of neighboring voxels allows higher statistical power, because adjacent voxels typically share contextual information from adjacent voxels.²¹ Such spatial correlations in the surrounding areas were useful for classification purposes using the contextual information from neighboring voxels.^{11,22–24} A recent study has shown that diffusion kurtosis imaging-derived scalar measures can be estimated from the significantly reduced number of diffusion encoding directions using a CNN algorithm.¹¹ To this end, accounting for the contextual information from the neighboring voxels could also be beneficial for estimating ATT and CBF, because the adjacent voxels in PWI may contain strongly correlated information. Thus, $3 \times 3 \times 3$ input patches for 3D CNN were extracted from each voxel to include its 26 neighboring voxels in all directions.

Based on the ASL standard kinetic model in Eq. (1), a quantitative CBF value depends on ATT, and the accuracy of ATT quantification affects CBF quantification.²⁵ However, ATT is independent of the quantitative CBF value, although a higher CBF, which yields a higher SNR, may improve the accuracy of ATT estimation. This is attributed to the fact that ATT can be independently estimated. ATT is based on the relative values in the perfusion time series, whereas CBF is based on various parameters including ATT.³ Depending on the ATT, which determines the position of the ASL kinetic curve in time, the amount

of T_1 relaxation and the magnitude of CBF vary.³ Therefore, a hierarchical network structure was implemented to reflect the model characteristics. The partially shared shallow layers and hierarchically structured deeper layers not only preserved the individual features between targets but accounted for common features. Taking these characteristics, ATT was estimated from the shallow layers, and CBF was estimated from the deeper layers. Through a partially shared network, the neural network can learn common features among related features of ATT and CBF, and specific features of CBF can be learned through the deeper layers.^{11,26,27} The network structures of standard CNN and H-CNN are illustrated in Figure 2. Note that all training weights were optimized using a gradient-based Adam optimizer,²⁸ which operates on the back-propagation.²⁹ To utilize both CNN and the nonlinear fitting method, we tested another estimation method, called a hybrid method, in which the estimations of ATT and CBF were divided into two steps. First, ATT was estimated from the standard CNN. Second, the estimated ATT map was used to estimate CBF using the nonlinear model fitting.

2.7 | Slice timing correction

In our 2D single-shot EPI acquisition of pCASL, each slice had a 27.5-ms slice delay per slice from the first slice to the most superior slice. For the nonlinear model fitting method, Eq. (1) accounts for the slice delays. Therefore, the neural networks also need to account for the slice delays. In the CNN models, a 3D timing delay map was accounted as an additional input feature. Each slice of the timing delay map contained the accumulated delay information from the first slice to the top slice of the brain.

2.8 | Experimental network settings and network optimization

Two differently structured CNNs, a standard CNN and H-CNN, were tested. Both networks included one input layer as a convolutional layer and several hidden layers. The convolutional layer containing $3 \times 3 \times 3$ input patches extracted input features and learned optimum spatial information. The hidden layers were fully connected layers with 1×1 kernels, which performed weighted linear recombination of the values from the previous layer.³⁰ The rectified linear unit activation³¹ was used in each hidden layer. Root mean square error (RMSE) was used for the loss function for both standard CNN and H-CNN.

To optimize the network structure, different numbers of hidden layers were trained and compared. The standard CNN was tested from two to six hidden layers. For H-CNN,

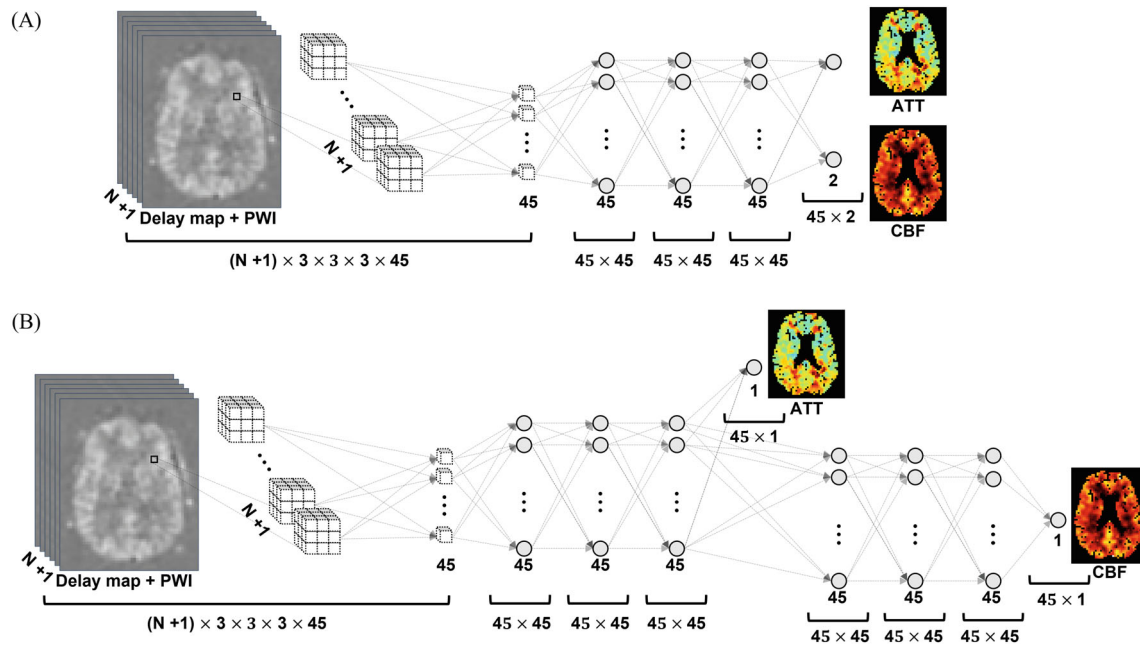


FIGURE 2 The structure of a standard convolutional neural network (CNN) (A) and a hierarchically structured 3D CNN (H-CNN) (B). $N+1$ denotes the number of PLDs and the delay map, which contains the slice timing information. Both networks included one input layer as a convolutional layer and multiple hidden layers. The numbers in each layer represent the numbers of neurons and weights.

the different combinations of the number of hidden layers were tested. The numbers of the shallow and deeper layers each varied from one to four independently. Different numbers of training subjects from 3 to 45 were also tested to investigate whether the numbers of training subjects affect the network training. To optimize the network training, initial learning rates were tested from 0.1 to 0.0001. The number of kernels in each hidden layer was also tested from 5 to 65. RMSE was used for the network optimization process. To improve the training performance, training was finished if the reduced training error in the 10th epoch exceeded a threshold of 0.0001.

2.9 | Training protocol and PLD selection

Forty-five randomly chosen subjects were used as training subjects. The training was performed within the brain mask, including gray and white matters, where the calculated ATT is less than 3962.5 ms because ATT estimation in the nonlinear fitting routine shows drastically large errors after 3000 ms, which is the longest PLD.⁴ The limit of 3962.5 ms includes the maximum slice timing delay, which is 962.6 ms at the top slice of the brain. In the training stage, the Xavier uniform initializer was used for all the trainable weights.³²

Two different data-reduction schemes were used for PLDs and averages. The PLD reduction scheme was every single combination of PLDs with the fixed six averages,

because each PLD can differently affect the estimation of the ASL standard model in both nonlinear model fitting and CNN. The reduction scheme of the number of averages was sequential selections. First, N averages were sequentially selected from six to three averages with the fixed six PLDs from the data set. Using only one or two averages was excluded in the experiments, because acquiring one or two images for long PLDs is not practical due to very low SNR. Every single network was trained individually when the numbers of PLDs or averages were reduced with the ground-truth reference images as targets for each subject. The combinations of the reduced number of PLDs were chosen from the lowest overall RMSEs of 3 test subjects from the nonlinear fitting and H-CNN separately. Due to the different physical values of ATT and CBF, overall RMSEs of ATT and CBF were normalized by their mean values to provide the best combinations of PLDs. In addition, an extreme case such as two PLDs and three averages was tested using the predetermined two PLDs that allowed the lowest overall RMSEs.

2.10 | Evaluation of predicted images

Mean absolute error (MAE) and RMSE were used as primary evaluation metrics to compare the estimated measures to their ground-truth reference images. Mean error (ME) was also used to compare the ground-truth

references to the estimated maps of ATT and CBF. Note that RMSE was used primarily for algorithm optimization, whereas MAE and ME were used to report the final performance of the algorithm. The MAE, RMSE, and ME were calculated over all brain voxels within the brain mask, including gray and white matters, using the following equations:

$$\text{MAE} = \frac{\sum_{i=1}^V |\hat{v}_i - v_{refi}|}{V}, \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^V (\hat{v}_i - v_{refi})^2}{V}}, \quad (3)$$

$$\text{ME} = \frac{\sum_{i=1}^V \hat{v}_i - v_{refi}}{V}, \quad (4)$$

where V is the total number of voxels; $\hat{v}_i$ is the estimated value at the i th voxel; v_{refi} is a reference value at the i th voxel; and $\hat{v}_i - v_{refi}$ is the estimation error of the i th voxel. The reference values are from the ATT and CBF maps of each subject using the nonlinear fitting method from the full data set. Additionally, difference maps were created by subtracting the estimated measures from the reference image to visually compare the estimated measures and their references. These difference maps allowed us to identify whether there were any tendencies of estimation errors by the location of the brain for both ATT and CBF.

2.11 | Test-retest reliability analysis

To assess the reproducibility of the H-CNN model, the test-retest reliability analysis was performed. Three years of follow-up scans of 4 subjects were acquired using identical acquisition parameters, and the same

pretrained weights were used in the H-CNN model. The within-subject coefficients of variations (wsCVs) were calculated as a ratio of the SD of the difference between baseline and follow-up scans to the mean of repeated measurements in the whole-brain gray matter. Note that the wsCV values were calculated with the estimated ATT and CBF maps from the nonlinear fitting and H-CNN, including the reduced PLDs. The predetermined PLD combinations were used to produce their ATT and CBF maps.

2.12 | Implementation

All experiments were performed under Linux OS with NVIDIA Tesla P40 GPU, 2 × Intel Xeon Gold 6242 2.8G processors, and 4 × 32 GB RDIMM. All neural networks were implemented in Python using the Keras with a Tensorflow backend.

3 | RESULTS

3.1 | Slice timing correction

The slice timing delay resulted in biased estimations in the neural networks. Figure 3A shows that the superior part of the brain tends to be underestimated, and the inferior part of the brain tends to be overestimated. This is because the top slice of the brain has a 962.5-ms delay compared with the first slice, and CNN trains the model with the lowest estimation errors by estimating the model from the slice that has the greatest number of voxels. This estimation error was corrected by the additional 3D input feature that contains timing delay information of each slice of the brain (Figure 3B,C).

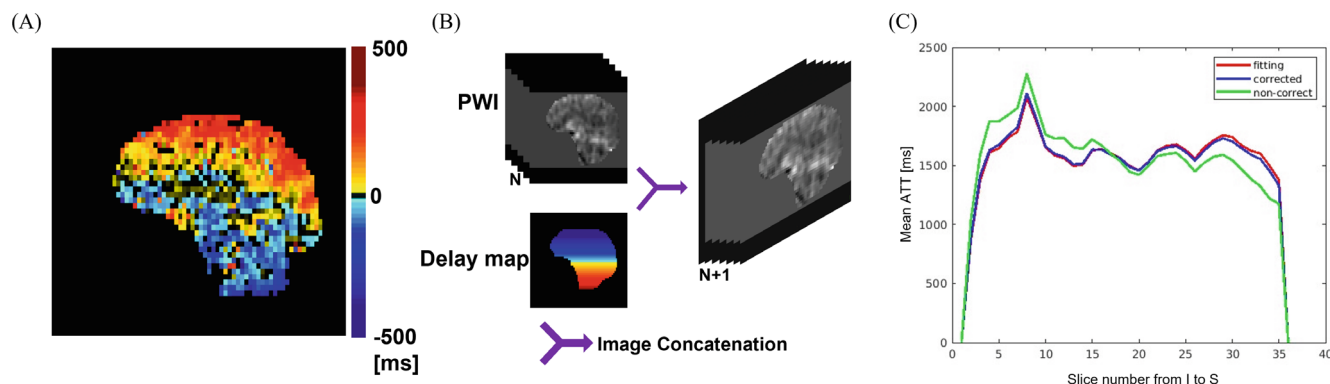


FIGURE 3 Estimation errors by the slice timing delay and corrections. (A) Difference map between the estimated arterial transit time (ATT) map and its reference. (B) A delay map as an additional input feature that corrects slice timing delay was concatenated into the input image. (C) Mean ATT plot. Each data point is the mean value of ATT of each brain slice from inferior to superior parts of the brain (reference image in red, non-slice timing-corrected estimation in green, and slice timing-corrected estimation in blue).

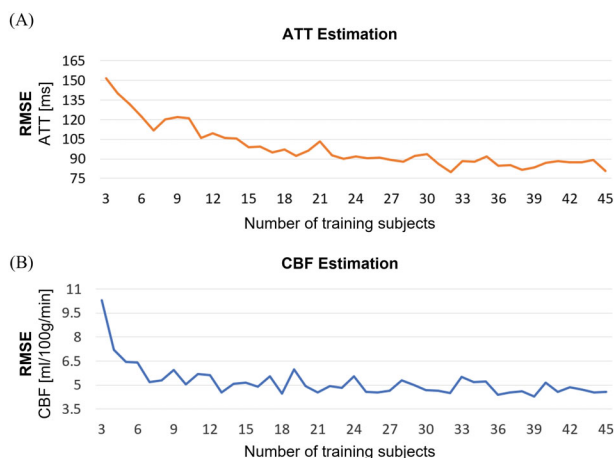


FIGURE 4 The overall RMS errors (RMSEs) of 3 test subjects with different numbers of training subjects: ATT estimation (A) and cerebral blood flow (CBF) estimation (B).

3.2 | Network optimization

From the network optimization process, an initial learning rate and the number of kernels in each hidden layer were chosen as 0.001 and 45, respectively. In addition, three hidden layers were chosen for the standard CNN. Three shallow layers and three deeper layers were also chosen for the H-CNN. Figure 4 shows the RMSE between ground-truth reference images and estimated maps by the different numbers of training subjects, which showed that the training was balanced after 35 training subjects.

3.3 | CNN

Both ATT and CBF estimation results were compared between the CNN and H-CNN (Figure 5). The overall

RMSEs showed that the CBF map was clearly estimated better in the H-CNN even though the difference in ATT estimation was not significant. Therefore, the results support that the hierarchical network structure with the additional hidden layers for CBF estimation in the H-CNN was beneficial. Additionally, the hybrid method showed that the estimation error of ATT from the CNN critically affected CBF estimation in the nonlinear model fitting (data not shown). Although the estimated ATT from the hybrid method was beneficial in estimating the CBF map compared with the nonlinear fitting method, the estimation error of the ATT map from the CNN part of the hybrid method affected CBF estimation in the nonlinear model fitting with higher RMSEs than the H-CNN. As a result, the H-CNN outperformed the hybrid method with respect to the overall MAEs and RMSEs. Consequently, further experiments were performed with the H-CNN.

3.4 | Reduced PLDs and averages

The conventional nonlinear model fitting method and H-CNN were compared with the reduced numbers of PLDs and averages. Figure 6 shows that the H-CNN outperforms the model fitting method when the number of PLDs or averages is reduced. The reduction of the number of averages produced larger errors on estimations of ATT and CBF than the reduction of PLDs. Figure 7 shows the estimated ATT and CBF maps of one of the test subjects from the nonlinear model fitting method and the H-CNN comparing the full data set with three PLDs or three averages. The rest of the estimated ATT and CBF maps of the subject using the different numbers of PLDs or averages are shown in Figures S1 and S2. The difference maps between the ground-truth references and estimated

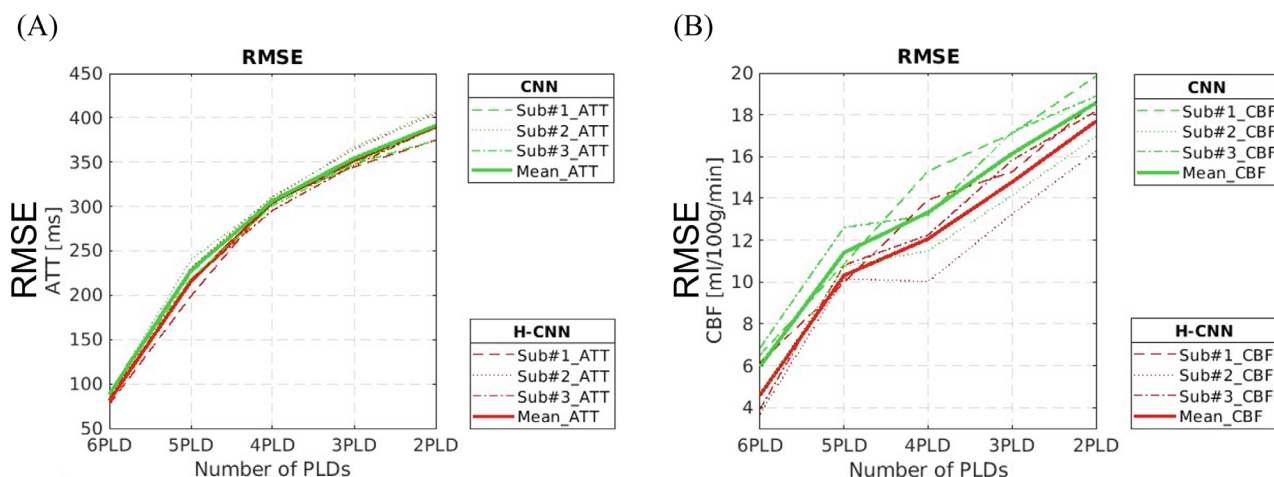


FIGURE 5 The overall RMSEs of the standard CNN (green) and H-CNN (red) with the reduced numbers of PLDs: ATT estimation (A) and CBF estimation (B).

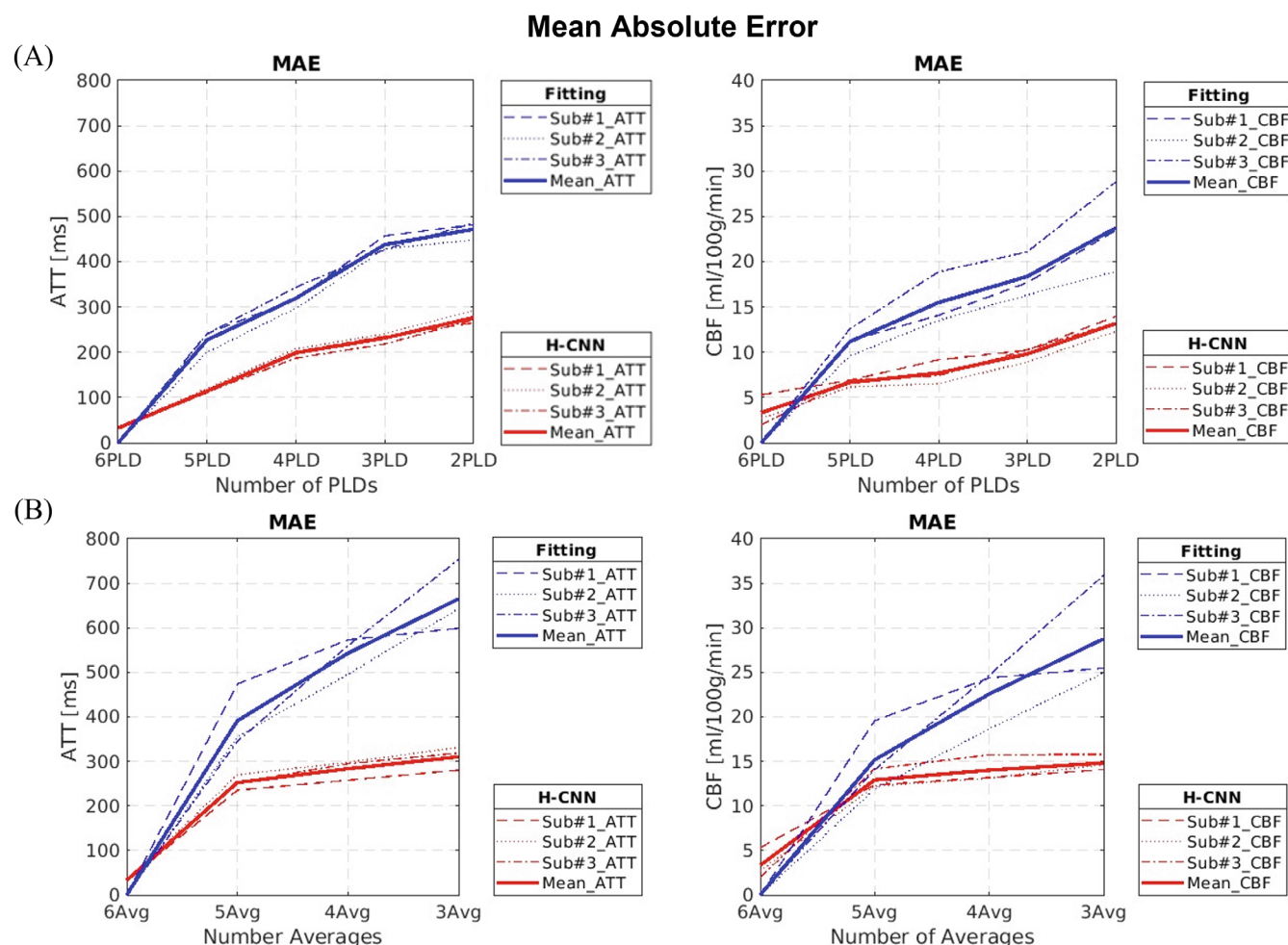


FIGURE 6 The overall mean absolute errors (MAEs) of the estimated ATT (left) and CBF (right) maps from the nonlinear fitting (blue) and H-CNN (red) using the reduced numbers of PLDs (A) and the reduced numbers of averages (B). The ATT and CBF maps from the nonlinear fitting with six PLDs and six averages were used for the ground-truth reference images for the MAE calculations.

images are shown in Figure S3. Figure S4 shows the estimated ATT and CBF maps and the MAEs of the extreme case that contains only two PLDs and three averages. The reported subject was selected due to the longest mean ATT among the 3 test subjects, which is more challenging to estimate ATT and CBF in practice. Total scan-time reduction by the reduced number of PLDs or averages in the H-CNN was calculated as the percent difference compared with the full data set (six PLDs and six averages), which was 5 min 21 s (Table 1). Also, the total scan time savings of each combination of PLDs are addressed in Table S1. Based on the selected PLDs (Table 1), the zero PLD (the first PLD) was necessary for better estimation. The ME and MAE between the ground-truth references and estimated maps are addressed in Table 2. When the number of PLDs was reduced, we observed that the discrepancies were not significant. Table 3 lists the mean values of the estimated ATT and CBF maps of all 3 test subjects, to address whether there are any subject-specific biases. All

3 test subjects showed that the H-CNN outperformed the nonlinear model fitting method.

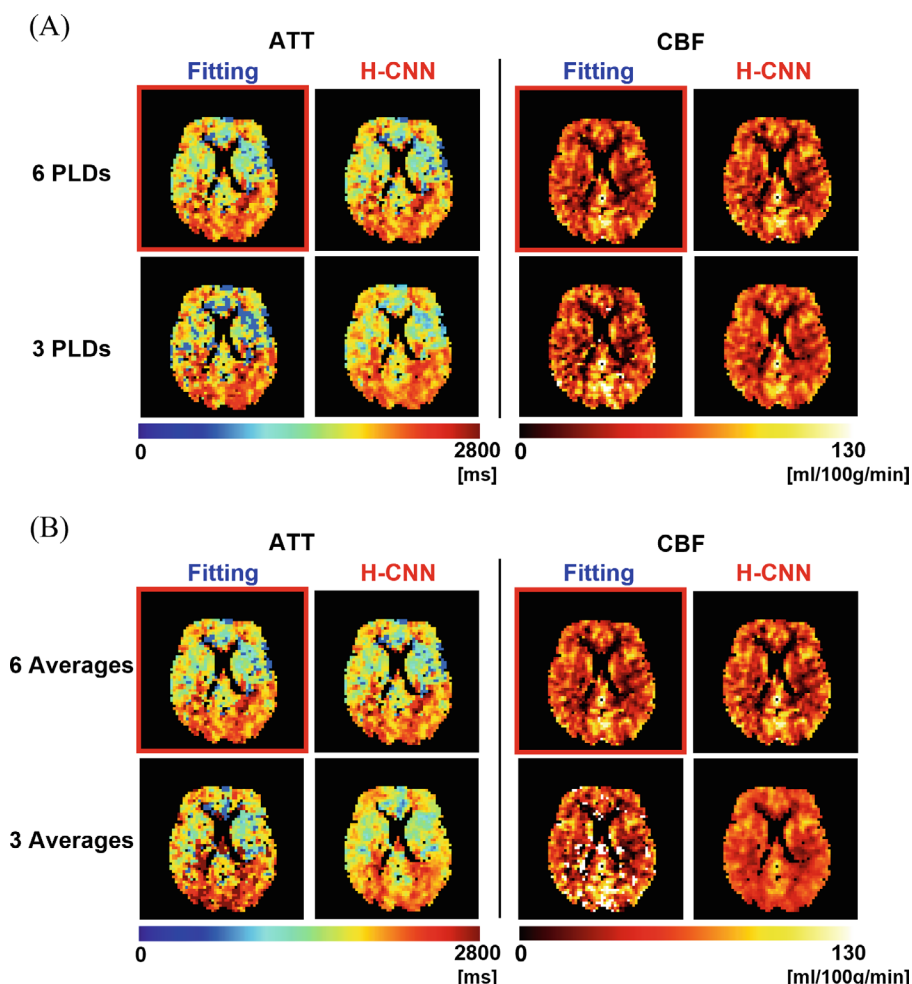
3.5 | Test-retest reliability analysis

Compared with the wsCV values from the nonlinear fitting methods (Table S2), the H-CNN showed higher reproducibility when the number of PLDs was reduced. The wsCV values are relatively high in both nonlinear fitting and H-CNN, but 3 years of the time interval between the baseline and follow-up scans might affect the wsCV (see Section 4).

4 | DISCUSSION

In this study, we demonstrated that the proposed H-CNN model can estimate ATT and CBF maps from multi-PLD

FIGURE 7 Estimated ATT and CBF maps from the nonlinear model fitting and H-CNN using the reduced numbers of PLDs (A) and averages (B). The ATT and CBF maps (red boxes) from the nonlinear model fitting with six PLDs and six averages are the ground-truth references of the subject.



pCASL PWI. We also showed that the reduced data, including reduced PLDs and reduced averages, can be recovered using the H-CNN. Compared with the H-CNN, the nonlinear model fitting method was dominantly affected by the reduced PLDs and averages (Figures 6 and 7). Thus, the H-CNN outperformed the conventional nonlinear model fitting method with the reduced number of PLDs or averages. The reported results showed that the smaller number of PLDs or averages can be used in the processing without significant discrepancy from the reference, which will allow a total scan-time reduction of the multi-PLD pCASL scheme.

Deep learning-based methods are most likely to require a large amount of data to train the network. We might need a larger amount of data, because ATT and CBF significantly vary by subject.^{4–6,33} CBF even varies within subjects due to various physiological factors such as PCO_2 .³³ In a recent deep learning-based study estimating ATT and CBF using U-Net and standard CNN, a total of 316 image sets including data augmentation were used.¹⁶ Additionally, the U-Net model outperformed CNN in their study. However, U-Net generally requires more training parameters, which also requires more training

TABLE 1 Time-saving by selected PLDs or averages for H-CNN.

Selected PLDs	Time Saving (Total Scan Time)
6 PLDs (all)	0% (5 min 21 s)
5 PLDs (1st, 2nd, 3rd, 4th, 6th)	20.08% (4 min 17 s)
4 PLDs (1st, 3rd, 4th, 5th)	35.61% (3 min 27 s)
3 PLDs (1st, 3rd, 4th)	55.68% (2 min 23 s)
2 PLDs (1st, 4th)	71.21% (1 min 32 s)
Number of Averages	Time Saving (Total Scan Time)
6	0% (5 min 21 s)
5	16.67% (4 min 28 s)
4	33.33% (3 min 24 s)
3	50.00% (2 min 41 s)
Extreme Case	Time Saving (Total Scan Time)
2 PLDs (1st, 4th) and 3 averages	85.61% (46 s)

TABLE 2 ME and MAE between the ground-truth references and estimated measures with the reduced numbers of PLDs and averages.

Number of PLDs	ME				MAE			
	Fitting		H-CNN		Fitting		H-CNN	
	ATT	CBF	ATT	CBF	ATT	CBF	ATT	CBF
6	0	0	1.46	0.47	0	0	32.69	3.32
5	2.36	-0.27	-2.66	0.51	227.02	11.16	114.18	6.66
4	8.98	-0.43	-3.01	0.34	319.28	15.48	198.69	7.70
3	9.73	-0.44	-5.10	0.11	437.44	18.36	231.45	9.80
2	24.01	-0.73	-0.20	0.24	471.61	23.73	276.39	13.17
Number of Averages	ATT	CBF	ATT	CBF	ATT	CBF	ATT	CBF
6	0	0	1.46	0.47	0	0	32.69	3.32
5	-2.84	0.32	-4.43	-0.04	390.89	15.18	252.26	12.91
4	-15.55	-1.10	-3.84	-0.06	542.81	22.55	283.40	14.01
3	-22.92	-1.87	0.44	0.31	664.82	28.76	310.16	14.82

Note: Each difference value indicates the ME or MAE across 3 test subjects (ATT in milliseconds and CBF in mL/100g/min). A full data set that contains six PLDs and six averages from the nonlinear model fitting was used for the ground-truth references.

TABLE 3 Mean of the estimated ATT and CBF maps of the 3 test subjects from the nonlinear model fitting and H-CNN.

PLDs	Mean of Estimated ATT and CBF											
	Test Subject #1				Test Subject #2				Test Subject #3			
	Fitting		H-CNN		Fitting		H-CNN		Fitting		H-CNN	
	ATT	CBF	ATT	CBF	ATT	CBF	ATT	CBF	ATT	CBF	ATT	CBF
6	1635.46	53.52	1631.75	50.38	1551.95	50.90	1549.35	48.59	1821.14	55.97	1805.05	57.15
5	1625.19	55.99	1674.12	51.10	1577.97	53.81	1563.64	48.27	1835.71	58.66	1806.75	55.95
4	1607.51	57.03	1649.42	50.68	1540.30	53.85	1573.37	49.66	1810.94	60.65	1827.06	57.36
3	1612.12	55.80	1688.09	52.98	1510.98	54.07	1596.19	51.34	1823.88	61.71	1788.91	56.50
2	1576.28	59.74	1640.71	51.87	1426.60	53.51	1574.43	50.32	1712.05	61.99	1789.91	56.97

Note: ATT is in milliseconds and CBF is in mL/100 g/min.

subjects.^{34,35} Therefore, using a CNN-based network structure is advantageous when the number of training data or computational power are limited. In our study, only 45 subjects were used for network training. Nonetheless, in this voxel-wise estimation, more than 27 000 samples were used from each subject to train the networks in the actual experiments. With this number of samples, the network was feasibly trained, and the results showed good performance. The total numbers of training parameters and samples were 18 992 and 1 255 590, respectively, for the model with the full data set.

Our study has several limitations that need to be addressed when interpreting the results. First, our ground-truth reference images might not be enough to be optimum gold-standard references. The optimum

gold-standard multi-PLD pCASL needs to be higher spatial resolution with higher SNR. The time to the maximum tissue residual function ($T_{\max}$) estimated with dynamic-susceptibility contrast MRI has been used to validate ATT because of its correlations with ATT.⁵ However, due to the lack of dynamic-susceptibility contrast data in this study, a further validation process would be needed to validate our ground-truth ATT. Moreover, in this study, there were limited selections of PLDs and the number of averages used. Although all possible combinations of PLDs were tested for the reduced numbers of PLDs in the H-CNN estimations, the training model only covered these specific PLDs acquired. If we had more PLDs with a smaller increment of the delay, the H-CNN could have provided better estimations. Based on the estimation errors

from the reduced averages compared with the reduced PLDs, the reduced averages deteriorated the estimation more than the reduced PLDs. Thus, ideally more averages would be acquired. Second, to create the reference image, $T_{1\text{tiss}}$ was assumed for a single T_1 value from existing literature for all brain tissue that is assumed to be the same as $T_{1\text{blood}}$.²⁰ This assumption led to an underestimation of both ATT and CBF, where the actual $T_{1\text{tiss}}$ is shorter than the assumed value.⁴ Third, partial volume (PV) effects were not corrected in this study. ASL has a relatively low spatial resolution, leading to an ASL perfusion map that is significantly affected by PV effects. This is because most brain voxels in the perfusion map are likely to contain signals from more than one tissue type.³⁶ However, in this study, PV correction was not applied, which resulted in smaller contrast differences between gray and white matters in both ATT and CBF maps. Fourth, a 3D pCASL sequence has not been tested. Because 3D ASL does not have a slice timing delay, the slice timing correction term will not be necessary. Fifth, training subjects were limited to the specific study group. Various factors such as age, hypertension, diabetes, and neurologic diseases can alter ATT and CBF.^{5,37–41} Pathological conditions could differently affect the estimation of both ATT and CBF maps in both the nonlinear model fitting and neural networks. Therefore, this study is limited to this specific study group. Finally, the test–retest reliability analysis was performed with the limited number of subjects ($n = 4$). Additionally, the 3-year time interval between baseline and follow-up scans may have contributed to the high wsCV values. This is because the reproducibility is affected by physiological fluctuations and repositioning errors.⁴²

Despite the addressed limitations, this study has the following potential clinical applications. This study showed that total scan time can be substantially reduced with the reduced number of PLDs or averages using the H-CNN. Additionally, using the reduced number of PLDs or averages can improve image quality. To be specific, when a part of the acquisition is motion-corrupted, the corrupted part can be recovered using the H-CNN. Moreover, this study may provide the best combination of PLDs if we can have more PLDs with smaller increments and a wider range of coverage of the labeling delays. In other words, this study may have the potential to provide the ideal choices of PLDs, which are independent of the nonlinear fitting method. This study can also be used to increase SNR. To be specific, once we have the H-CNN training weights from more averages, we can have a better SNR when we collect smaller numbers of averages. With this approach, higher spatial resolution can also be achieved in a shorter scan time. Once we have the data set using higher-resolution multishot acquisitions such as 3D gradient and spin echo instead of single-shot 2D EPI, the images

with higher resolution can be obtained within a shorter scan time.

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DATA AVAILABILITY STATEMENT

The patient data will not be shared directly due to the ethics of the clinical study at Wake Forest University. However, the source codes and pretrained weights are publicly available at https://github.com/UcdIrisLab/ASL_CNN.

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REFERENCES

1. Wolf RL, Detre JA. Clinical neuroimaging using arterial spin-labeled perfusion magnetic resonance imaging. *Neurotherapeutics*. 2007;4:346–359.
2. Gevers S, Majoie C, Van den Tweel X, Lavini C, Nederveen A. Acquisition time and reproducibility of continuous arterial spin-labeling perfusion imaging at 3T. *Am J Neuroradiol*. 2009;30:968–971.
3. Buxton RB. *Introduction to Functional Magnetic Resonance Imaging: Principles and Techniques*. Cambridge University Press; 2009.
4. Johnston ME, Lu K, Maldjian JA, Jung Y. Multi-TI arterial spin labeling MRI with variable TR and bolus duration for cerebral blood flow and arterial transit time mapping. *IEEE Trans Med Imaging*. 2015;34:1392–1402.
5. Wang DJ, Alger JR, Qiao JX, et al. Multi-delay multi-parametric arterial spin-labeled perfusion MRI in acute ischemic stroke—comparison with dynamic susceptibility contrast enhanced perfusion imaging. *NeuroImage Clin*. 2013;3:1–7.
6. Liu Y, Zhu X, Feinberg D, et al. Arterial spin labeling MRI study of age and gender effects on brain perfusion hemodynamics. *Magn Reson Med*. 2012;68:912–922.
7. Wang S, Jiang Y, Hou X, Cheng H, Du S. Cerebral micro-bleed detection based on the convolution neural network with rank based average pooling. *IEEE Access*. 2017;5:16576–16583.
8. Wu N, Phang J, Park J, et al. Deep neural networks improve radiologists' performance in breast cancer screening. *IEEE Trans Med Imaging*. 2019;39:1184–1194.
9. Hussein S, Kandel P, Bolan CW, Wallace MB, Bagci U. Lung and pancreatic tumor characterization in the deep learning era: novel supervised and unsupervised learning approaches. *IEEE Trans Med Imaging*. 2019;38:1777–1787.
10. Hansen B, Lund TE, Sangill R, Jespersen SN. Experimentally and computationally fast method for estimation of a mean kurtosis. *Magn Reson Med*. 2013;69:1754–1760.
11. Li Z, Gong T, Lin Z, et al. Fast and robust diffusion kurtosis parametric mapping using a three-dimensional convolutional neural network. *IEEE Access*. 2019;7:71398–71411.

12. Xie D, Li Y, Yang H, et al. Denoising arterial spin labeling perfusion MRI with deep machine learning. *Magn Reson Imaging*. 2020;68:95-105.
13. Zhang L, Xie D, Li Y, et al. Improving sensitivity of arterial spin labeling perfusion MRI in Alzheimer's disease using transfer learning of deep learning-based ASL denoising. *J Magn Reson Imaging*. 2021;55:1710-1722.
14. Gong K, Han P, El Fakhri G, Ma C, Li Q. Arterial spin labeling MR image denoising and reconstruction using unsupervised deep learning. *NMR Biomed*. 2019;35:e4224.
15. Kim KH, Choi SH, Park S-H. Improving arterial spin labeling by using deep learning. *Radiology*. 2018;287:658-666.
16. Luciw NJ, Shirzadi Z, Black SE, Goubran M, MacIntosh BJ. Automated generation of cerebral blood flow and arterial transit time maps from multiple delay arterial spin-labeled MRI. *Magn Reson Med*. 2022;88:406-417.
17. Penny WD, Friston KJ, Ashburner JT, Kiebel SJ, Nichols TE. *Statistical Parametric Mapping: The Analysis of Functional Brain Images*. Elsevier; 2011.
18. Buxton RB, Frank LR, Wong EC, Siewert B, Warach S, Edelman RR. A general kinetic model for quantitative perfusion imaging with arterial spin labeling. *Magn Reson Med*. 1998;40:383-396.
19. Çavuşoğlu M, Pfeuffer J, Uğurbil K, Uludağ K. Comparison of pulsed arterial spin labeling encoding schemes and absolute perfusion quantification. *Magn Reson Imaging*. 2009;27:1039-1045.
20. Lu H, Clingman C, Golay X, Van Zijl PC. Determining the longitudinal relaxation time (T_1) of blood at 3.0 tesla. *Magn Reson Med*. 2004;52:679-682.
21. Yuan L, Wei X, Shen H, Zeng L-L, Hu D. Multi-center brain imaging classification using a novel 3D CNN approach. *IEEE Access*. 2018;6:49925-49934.
22. Pereira S, Pinto A, Alves V, Silva CA. Brain tumor segmentation using convolutional neural networks in MRI images. *IEEE Trans Med Imaging*. 2016;35:1240-1251.
23. Zou L, Zheng J, Miao C, Mckeown MJ, Wang ZJ. 3D CNN based automatic diagnosis of attention deficit hyperactivity disorder using functional and structural MRI. *IEEE Access*. 2017;5:23626-23636.
24. Liu Z, Cao C, Ding S, Liu Z, Han T, Liu S. Towards clinical diagnosis: automated stroke lesion segmentation on multi-spectral MR image using convolutional neural network. *IEEE Access*. 2018;6:57006-57016.
25. Alsop DC, Detre JA. Reduced transit-time sensitivity in noninvasive magnetic resonance imaging of human cerebral blood flow. *J Cereb Blood Flow Metab*. 1996;16:1236-1249.
26. Caruana R. *Learning many related tasks at the same time with backpropagation*. In *Proceedings of the NeurIPS*. Denver, Colorado, USA; 1995:657-664.
27. Ghosh J, Bengio Y. Multi-task learning for stock selection. In *Proceedings of the NeurIPS*. Denver, Colorado, USA; 1997:946-952.
28. Kingma DP, Ba J. Adam: a method for stochastic optimization. *arXiv Preprint*. arXiv:1412.6980 2014.
29. LeCun Y, Boser B, Denker JS, et al. Backpropagation applied to handwritten zip code recognition. *Neural Comput*. 1989;1:541-551.
30. Lin M, Chen Q, Yan S. Network in network. *arXiv Preprint*. arXiv:1312.4400 2013.
31. Glorot X, Bordes A, Bengio Y. Deep sparse rectifier neural networks. In *Proceedings of the JMLR Workshop and Conference*, Budapest, Hungary, 2011. pp. 315-323.
32. Glorot X, Bengio Y. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the JMLR Workshop and Conference*, Sardinia, Italy, 2010. pp. 249-249.
33. Henriksen OM, Kruuse C, Olesen J, et al. Sources of variability of resting cerebral blood flow in healthy subjects: a study using ^{133}Xe SPECT measurements. *J Cereb Blood Flow Meta*. 2013;33:787-792.
34. Minnema J, Wolff J, Koivisto J, et al. Comparison of convolutional neural network training strategies for cone-beam CT image segmentation. *Comput Methods Programs Biomed*. 2021;207:106192.
35. Ghodrati V, Shao J, Bydder M, et al. MR image reconstruction using deep learning: evaluation of network structure and loss functions. *Quant Imaging Med Surg*. 2019;9:1516-1527.
36. Asllani I, Borogovac A, Brown TR. Regression algorithm correcting for partial volume effects in arterial spin labeling MRI. *Magn Reson Med*. 2008;60:1362-1371.
37. Kim D, Hughes TM, Lipford ME, et al. Relationship between cerebrovascular reactivity and cognition among people with risk of cognitive decline. *Front Physiol*. 2021;12:645342.
38. Tsuda Y, Hartmann A. Changes in hyperfrontality of cerebral blood flow and carbon dioxide reactivity with age. *Stroke*. 1989;20:1667-1673.
39. Bracko O, Njiru BN, Swallow M, Ali M, Haft-Javaherian M, Schaffer CB. Increasing cerebral blood flow improves cognition into late stages in Alzheimer's disease mice. *J Cereb Blood Flow Metab*. 2020;40:1441-1452.
40. Muller M, van der Graaf Y, Visseren FL, Mali WPTM, Geerlings MI, SMART Study Group. Hypertension and longitudinal changes in cerebral blood flow: the SMART-MR study. *Ann Neurol*. 2012;71:825-833.
41. Gonzalez-At JB, Alsop DC, Detre JA. Cerebral perfusion and arterial transit time changes during task activation determined with continuous arterial spin labeling. *Magn Reson Med*. 2000;43:739-746.
42. Chen Y, Wang DJ, Detre JA. Test-retest reliability of arterial spin labeling with common labeling strategies. *J Magn Reson Imaging*. 2011;33:940-949.

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

Figure S1. Estimated arterial transit time (ATT) and cerebral blood flow (CBF) maps from the nonlinear model fitting and hierarchically structured 3D convolutional neural network (H-CNN) using the reduced numbers of postlabeling delays (PLDs). The ATT and CBF maps (red boxes) from the nonlinear model fitting with six PLDs and six averages are the ground-truth references of the subject.

Figure S2. Estimated ATT and CBF maps from the nonlinear model fitting and H-CNN using the reduced numbers of averages. The ATT and CBF maps (red boxes) from the

nonlinear model fitting with six PLDs and six averages are the ground-truth references of the subject.

Figure S3. The difference maps between the ground-truth references and estimated images from the nonlinear fitting and H-CNN using the reduced numbers of PLDs (A) and averages (B).

Figure S4. Estimated ATT and CBF maps from the nonlinear model fitting (left) and H-CNN (right) using the extreme case containing two PLDs and three averages.

Table S1. Time-saving by every combination of PLDs.

Table S2. The within-subject coefficients of variation (wsCV) values of 4 subjects from the nonlinear fitting and H-CNN.

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